

Physiology Practice Questions: DM Module

Lecture Title: Functions and Regulation of the GI Tract (L20)

Note: If you need any clarification, please email to: ASK-PHY@squ.edu or email to the faculty who delivered this lecture.

1. A 37-year-old woman comes to the physician because of a 4-month history of recurrent abdominal pain in the right upper quadrant. Her BMI is 36 kg/m² (Normal: 19-25). Episodes occur a few times each month, last 1 to 2 hours, and cause nausea and sometimes vomiting. Laboratory studies show elevated serum concentration of the hormone, cholecystokinin during these episodes. Which of the following most likely stimulates this hormone secretion?

- A. Peristalsis in the esophagus by swallowing
- B. Sympathetic nerve stimulation
- C. Fatty acids and amino acids in the duodenum
- D. Distension of the stomach
- E. Gastric acid secretion

2. A 53-year-old woman comes to the clinic because of a 6-week history of profuse diarrhea. He also has muscle weakness. Physical examination shows dry mucous membranes and reduced skin turgor. Laboratory studies show elevated serum concentration of vasointestinal peptide (VIP). CT scan of the abdomen shows a tumor in the pancreas that secretes this hormone. This hormone is physiologically secreted from the gut as one of the gut hormones. Which of the following physiological effects is most likely of this hormone?

- A. Vasoconstriction of blood vessels
- B. Inhibits bicarbonate secretion from the pancreas
- C. Stimulates acid secretion from stomach
- D. Relaxation of intestinal smooth muscles
- E. Inhibits intestinal water secretion

3. A 44-year-old man comes to the physician because of a 5-month history of three episodes of intermittent right upper quadrant abdominal pain. He says the pain occurs after eating heavy meals. The pain radiates to the right shoulder. Physical examination shows obesity. Abdominal ultrasonography shows gall stones. He is diagnosed with gall stone disease. The pain following a heavy meal is because of release of gut hormone. Which of the following physiological effects is most likely of this hormone?

- A. Inhibits gastric emptying
- B. Inhibits pancreatic bicarbonate secretion
- C. Inhibits pancreatic enzyme secretion

- D. Inhibits contraction of the gall bladder
- E. Stimulates contraction of the sphincter of Oddi

4. A 41-year-old man comes to the physician for a routine examination. He says that he has increased thirst and appetite for the past few months. Both of his parents have type 2 diabetes mellitus. Physical examination shows no abnormalities. Laboratory studies show borderline elevation in random serum glucose concentration. A glucose tolerance test is performed. Following ingestion of glucose increases serum concentration of a gut hormone in the patient. Which of the following gut hormones is most likely secreted in response to an oral glucose load in this patient?

- A. Secretin
- B. Gastrin
- C. CCK
- D. Vasoactive intestinal peptide
- E. Glucose-dependent insulinitropic peptide

5. Investigators conduct a study related to the intestinal motility *in vivo* using rat small intestine. Stimulation of a nerve within the enteric nervous system causes contraction of intestinal smooth muscle cells. Which of the following neurotransmitters is most likely released by this nerve?

- A. Acetylcholine (ACh)
- B. Vasoactive intestinal polypeptide (VIP)
- C. Nitric oxide (NO)
- D. Adenosine triphosphate (ATP)
- E. Somatostatin (SST)

6. Investigators conduct a study that evaluates the effect of various factors on secretion of gut hormones. Stomach of intact, anesthetized dog is used for this purpose. A solution with low pH (acidic) is placed on the antral region of the dog's stomach and serum concentrations of various gut hormones is estimated. It is observed that the low pH inhibited secretion of one of the gut hormones. Which of the gut hormones is most likely?

- A. Secretin
- B. Gastrin
- C. CCK
- D. VIP
- E. Gastric inhibitory peptide

7. A 34-year-old woman comes to the physician for a routine examination. She is very conscious of her body weight and figure. She says she takes injection of a synthetic analog of GLP-1 (glucagon-like peptide-1) to prevent obesity. Her BMI is 23 kg/m²

(Normal: 19-25). Physical examination shows no abnormalities. She is advised to stop taking this medication and encouraged to work out. GLP-1 is also secreted from the small intestine as one of the gut hormones. Which of the following sets of features best matches this gut hormone?

Option	Cell source in small intestine	Stimulus for release	Physiological effect
A	S cells	Glucose in stomach	Increases appetite
B	K cells	Glucose meal	Inhibits insulin secretion
C	L cells	Nutrients in small intestine	Inhibits appetite
D	G cells	Acid in the stomach	Inhibits acid secretion
E	D cells	Distention of stomach	Reduces gastric emptying

8. A 27-year-old man is brought to the emergency department because of a 2-day history of refusal to take food and fluids. He is lethargic and severely dehydrated. His pulse is 100/min and blood pressure is 110/64 mm Hg. Physical examination shows sunken eyes, dry mouth and tongue, and low skin turgor. Which of the following gastrointestinal regulatory substances is most likely to increase in this patient's serum?

- A. Gastrin
- B. Somatostatin
- C. Glucose dependent insulintropic peptide
- D. Cholecystokinin
- E. Ghrelin

Answer Key with Explanation

1. A 37-year-old woman comes to the physician because of a 4-month history of recurrent abdominal pain in the right upper quadrant. Her BMI is 36 kg/m² (Normal: 19-25). Episodes occur a few times each month, last 1 to 2 hours, and cause nausea and sometimes vomiting. Laboratory studies show elevated serum concentration of the hormone, cholecystokinin during these episodes. Which of the following most likely stimulates this hormone secretion?

- A. Peristalsis in the esophagus by swallowing
- B. Sympathetic nerve stimulation
- C. Fatty acids and amino acids in the duodenum
- D. Distension of the stomach
- E. Gastric acid secretion

Answer Key: C

Rationale: The key stimulus for CCK release is the presence of amino acids and fatty acids in the duodenum and proximal small bowel. These substrates stimulate the I cells present in those locations (mostly in the duodenum) to release CCK. The responsiveness of CCK release to these food substrates makes it the key mediator between meal composition and digestive response. The result is slower digestion and more enzyme and bile release in response to meals that contain more fat and protein.

2. A 53-year-old woman comes to the clinic because of a 6-week history of profuse diarrhea. He also has muscle weakness. Physical examination shows dry mucous membranes and reduced skin turgor. Laboratory studies show elevated serum concentration of Vaso intestinal peptide (VIP). CT scan of the abdomen shows a tumor in the pancreas that secretes this hormone. This hormone is physiologically secreted from the gut as one of the gut hormones. Which of the following physiological effects is most likely of this hormone?

- A. Vasoconstriction of blood vessels
- B. Inhibits bicarbonate secretion from the pancreas
- C. Stimulates acid secretion from stomach
- D. Relaxation of intestinal smooth muscles
- E. Inhibits intestinal water secretion

Answer Key: D

Rationale: VIP is isolated from the gut as a gut hormone. It is produced by enteric neurons throughout the GI tract and acts in a paracrine fashion (diffusion directly onto target cells through interstitial space) to induce smooth muscle relaxation. It also acts in endocrine fashion (transported to site of action via blood). It relaxes smooth muscles in blood vessels, stimulates pancreatic secretion of bicarbonate, and stimulates intestinal secretion of water and electrolytes. Hypersecretion of this hormone produces watery (secretory) diarrhea.

3. A 44-year-old man comes to the physician because of a 5-month history of three episodes of intermittent right upper quadrant abdominal pain. He says the pain occurs after eating heavy meals. The pain radiates to the right shoulder. Physical examination shows obesity. Abdominal ultrasonography shows gall stones. He is diagnosed with gall stone disease. The pain following a heavy meal is because of release of gut hormone. Which of the following physiological effects is most likely of this hormone?

- A. Inhibits gastric emptying
- B. Inhibits pancreatic bicarbonate secretion
- C. Inhibits pancreatic enzyme secretion
- D. Inhibits contraction of the gall bladder
- E. Stimulates contraction of the sphincter of Oddi

Answer Key: A

Rationale: CCK slows gastric motility and emptying. This provides sufficient time for digestion and absorption of protein and fat in the meal. CCK potentiates pancreatic enzyme and bicarbonate secretion and causes contraction of the gall bladder and relaxation of the sphincter of Oddi.

4. A 41-year-old man comes to the physician for a routine examination. He says that he has increased thirst and appetite for the past few months. Both of his parents have type 2 diabetes mellitus. Physical examination shows no abnormalities. Laboratory studies show borderline elevation in random serum glucose concentration. A glucose tolerance test is performed. Following ingestion of glucose increases serum concentration of a gut hormone in the patient. Which of the following gut hormones is most likely secreted in response to an oral glucose load in this patient?

- A. Secretin
- B. Gastrin
- C. CCK
- D. Vasoactive intestinal peptide
- E. Glucose-dependent insulinitropic peptide

Answer Key: E

Rationale: Glucose-dependent insulinitropic peptide (also known as Gastric inhibitory peptide, GIP) is secreted in response to glucose ingestion. Another hormone, GLP-1 (Glucagon-like peptide-1) is also released in response to oral glucose load. These 2 are known as incretins. Secretion of the other mediators listed as answer options is not glucose dependent.

5. Investigators conduct a study related to the intestinal motility *in vivo* using rat small intestine. Stimulation of a nerve within the enteric nervous system causes contraction of intestinal smooth muscle cells. Which of the following neurotransmitters is most likely released by this nerve?

- A. Acetylcholine (ACh)
- B. Vasoactive intestinal polypeptide (VIP)
- C. Nitric oxide (NO)
- D. Adenosine triphosphate (ATP)
- E. Somatostatin (SST)

Answer Key: A

Rationale: Both the parasympathetic and enteric nerves release ACh from their terminals in the GIT, which causes intestinal smooth muscle contraction. VIP, NO and ATP are also released by enteric neurons, but cause smooth muscle relaxation. SST is primarily from non-neuronal sources (D cells in the gastric and intestinal mucosa and islets of the pancreas), although it is also produced by enteric neurons in small amounts. However, it would cause smooth muscle relaxation rather than contraction.

6. Investigators conduct a study that evaluates the effect of various factors on secretion of gut hormones. Stomach of intact, anesthetized dog is used for this purpose. A solution with low pH (acidic) is placed on the antral region of the dog's stomach and serum concentrations of various gut hormones is estimated. It is observed that the low pH inhibited secretion of one of the gut hormones. Which of the gut hormones is most likely?

- A. Secretin
- B. Gastrin
- C. CCK
- D. VIP
- E. Gastric inhibitory peptide

Answer Key: B

Rationale: Gastrin is the key stimulating mediator for gastric acid release from parietal cells in the stomach. To prevent overly strong acidification of the stomach, a negative feedback loop is in place whereby G cells, which produce gastrin, are inhibited by strong acidity in the lumen of the stomach, especially near the antrum where most of the G cells are located. Gastrin production is inhibited by gastric luminal acidity via this feedback loop.

7. A 34-year-old woman comes to the physician for a routine examination. She is very conscious of her body weight and figure. She says she takes injection of a synthetic analog of GLP-1 (glucagon-like peptide-1) to prevent obesity. Her BMI is 23 kg/m² (Normal: 19-25). Physical examination shows no abnormalities. She is advised to stop taking this medication and encouraged to work out. GLP-1 is also secreted from the small intestine as one of the gut hormones. Which of the following sets of features best matches this gut hormone?

Option	Cell source in small intestine	Stimulus for release	Function
A	S cells	Glucose in stomach	Increases appetite
B	K cells	Glucose meal	Inhibits insulin secretion
C	L cells	Nutrients in small intestine	Inhibits appetite
D	G cells	Acid in the stomach	Inhibits acid secretion
E	D cells	Distention of stomach	Reduces gastric emptying

Answer Key: D.

Rationale: GLP-1 and GIP are considered as incretins. Both are secreted from the small intestine in response to eating. Both increase insulin secretion from the pancreas. GLP-1 is secreted from the L cells of the ileum in response to the presence of nutrients in the small intestine. Physiological effects of GLP-1 include stimulation of insulin secretion, inhibition of glucagon secretion, slowing gastric emptying, and reducing appetite.

Because of these functions, its analog drugs are used to treat obesity and reduce blood glucose concentration.

8. A 27-year-old man is brought to the emergency department because of a 2-day history of refusal to take food and fluids. He is lethargic and severely dehydrated. His pulse is 100/min and blood pressure is 110/64 mm Hg. Physical examination shows sunken eyes, dry mouth and tongue, and low skin turgor. Which of the following gastrointestinal regulatory substances is most likely to increase in this patient's serum?

- A. Gastrin
- B. Somatostatin
- C. Glucose dependent insulintropic peptide
- D. Cholecystokinin
- E. Ghrelin

Answer Key: E.

Rationale: This patient is in fasting state. Two major gastrointestinal regulatory substances secreted during fasting are Ghrelin and Motilin. Ghrelin is secreted from stomach and motilin from the small intestine. Ghrelin increases appetite.